Exploring GenAI-Driven Personalized Health Coaching for Chronic Disease Self-Management

Chronic diseases are a growing challenge in Singapore, with one in four adults aged 40 and above living with at least one condition. Providing continuous, personalised support outside clinical settings remains difficult, contributing to poor medication adherence, lifestyle barriers, and reduced patient engagement.
This project aims to enhance chronic disease self-management by using clinical text mining and Large Language Models (LLMs) to analyze patients' symptom and behaviour trajectories. By extracting insights from unstructured clinical and patient-generated text, we can track changes over time, identify patterns, and detect early signs of disease progression or complications. These insights support proactive, timely interventions and ultimately improve patient outcomes.

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A key question is how to bridge the gap between patients and clinicians: How can insights generated by the AI health companion be summarised and presented in a clear, clinically meaningful way—enabling earlier, data-driven interventions without adding to clinicians' administrative workload?